In [2]:
```
import pandas as pd
from sklearn.model_selection import train_
from sklearn.linear_model import LinearReg
from sklearn.metrics import r2_score
```

In [3]:
```
df = pd.read_csv("ratings.tsv", sep="\t")
df.head()
```

Out[3]:

<style scoped> .dataframe tbody tr th:only-of-type { vertical-align: middle; } .dataframe tbody tr th { vertical-align: top; } .dataframe thead th { text-align: right; } </style> <thead> <th></th> <th>restaurant</th> <th>food</th> <th>ambience</th> <th>service</th> <th>rating</th> </thead> <tbody> <th>0</th> <th>1</th> <th>2</th> <th>3</th> <th>4</th> </tbody> <tbody></tbody>

1	85	82	89	78	
2	80	90	80	85	
3	83	86	83	85	
4	70	96	75	72	
5	68	80	78	75	

In [4]:
```
print(df.columns)
print(df.shape)
```

```
Index(['restaurant', 'food', 'ambience', 's
ervice', 'rating'], dtype='str')
(15, 5)
```

In [5]:
```
X = df[["ambience", "food", "service"]]
y = df["rating"]
```

```
print(X.head())
print(y.head())
```

```
   ambience  food  service
0        82    85       89
1        90    80       80
2        86    83       83
3        96    70       75
4        80    68       78
0    78
1    85
2    85
3    72
4    75
Name: rating, dtype: int64
```

In [6]:
```
X_train, X_test, y_train, y_test = train_t
    X, y, test_size=0.2, random_state=42
)

print("Training rows:", len(X_train))
print("Testing rows:", len(X_test))
```

```
Training rows: 12
Testing rows: 3
```

In [7]:
```
model = LinearRegression()
model.fit(X_train, y_train)

print("Model trained!")
```

```
Model trained!
```

In [8]:
```
print("Ambience coefficient:", model.coef_
print("Food coefficient:", model.coef_[1])
print("Service coefficient:", model.coef_[
```

```
Ambience coefficient: 0.07817059371460455
Food coefficient: 0.8802155056349206
Service coefficient: 0.44168248839050595
```

In [9]:
```
y_pred = model.predict(X_test)

print("Actual ratings:", list(y_test))
print("Predicted ratings:", y_pred)
print("R-squared score:", r2_score(y_test,
```

```
Actual ratings: [77, 76, 78]
Predicted ratings: [75.11153264 74.57763953
90.25358372]
R-squared score: -76.86986617467137
```

which factor should a struggling restaurant fix first? After I ranked coefficents,I got this result (Ambience coefficient: 0.07817059371460455) (Food coefficient: 0.8802155056349206)(Service coefficient: 0.44168248839050595) .The restaurant should investigate improving the food quilty first because it get the highest coefficient result.followed by the service and ambience.however,after i got a negative R squared which is -76.87 its indicating a very poor presictive performance on the three test of the restaurant .the owner should treat this kind of ranking as a prelimnary and start collecting more data before making major decisions.

In [ ]: